Project List

1 Project Requirements

- Each group is suggested to choose one of the projects listed below.
- Some projects may have specific requirements. For the other projects that only have links of the paper, you are required to reproduce the results. Reference to the original code on Github is encouraged. But you should write your own code.
- The files to be submitted include code, a short description of results and slides that take approximately 10 minutes.
- Extra credit is available for the projects that have any one of the following properties.
  - Modify the existing model structure or propose a new model structure. Prove it has better performance than the model in the paper.
  - Have very deep exploration or investigation on the topic.
- You can also propose your own topic if you have some interesting ideas. But the approval is needed for new topics.

2 List of Projects

Project 1: Deep Q learning

All the papers below are Deep Q Learning rated from the original one to the complicated one. The goal of this project is to explore the DQN reinforcement learning. An ideal path is to go over all the papers from the list and write a brief summary of each and implement a couple with some well-known environment or your own designed environment.

Playing Atari with Deep Reinforcement Learning, Mnih et al, 2013. Algorithm: DQN.
Link: https://www.cs.toronto.edu/~vmnih/docs/dqn.pdf

Deep Recurrent Q-Learning for Partially Observable MDPs, Hausknecht and Stone, 2015. Algorithm: Deep Recurrent Q-Learning.
Link: https://arxiv.org/abs/1507.06527

Deep Reinforcement Learning with Double Q-learning, Hasselt et al 2015. Algorithm: Double DQN.

Link: https://arxiv.org/abs/1509.06461

Dueling Network Architectures for Deep Reinforcement Learning, Ziyu Wang etc.
Link: https://arxiv.org/pdf/1511.06581 Prioritized Experience Replay,Tom Schaul, John Quan, Ioannis Antonoglou, David Silver

Link: https://arxiv.org/pdf/1511.05952
Rainbow: Combining Improvements in Deep Reinforcement Learning, Hessel et al, 2017. Link: https://arxiv.org/abs/1710.02298

Project 2: Inventory decision-making

We manage to obtain the sales data of around 100 products for around 60 days in an e-commerce company. Some attributes of products, customers and the day. We have done the preprocessing of the data and split the data into the training and test data set. We provide complete information on the training table but hide the column 'sales' on the test table. Note that these products belong to the same category. You need to predict the sales on the test table. Moreover, you would need to make the inventory decision. The performance of your project depends on the prediction accuracy and the profit induced by your decision. The data will be released in person.

Project 3: Steel Plates Faults Multi-Classification Modeling

In this project, you will need to develop a machine learning model to identify seven types of defects that occur during the steel plate manufacturing process. The detailed description of the dataset will be released in person.

Project 4: Meta-Learning

All the papers below are Meta-Learning rated. The goal of this project is to explore Optimization-based and Metric-based Meta-Learning.

Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks, Finn et al, 2017. Algorithm: MAML.
Link: https://arxiv.org/abs/1703.03400

On First-Order Meta-Learning Algorithms Alex Nichol, Joshua Achiam, John Schulman.
Link: https://arxiv.org/pdf/1803.02999.pdf

Matching Networks for One Shot Learning.
Link: http://papers.nips.cc/paper/6385-matching-networks-for-one-shot-learning. pdf

Prototypical Networks for Few-shot Learning
Link: https://arxiv.org/pdf/1703.05175.pdf

Project 5: Convolutional Neural Networks for Text Classification.

Implement the following two papers:

- Very Deep Convolutional Networks for Text Classification Link: https://arxiv.org/pdf/1606.01781.pdf
- Densely Connected Convolutional Networks
  Link: https://openaccess.thecvf.com/content_cvpr_2017/papers/Huang_Densely_Connected_Convolutional_CVPR_2017_paper.pdf

Requirements:

- Implement the models in the above two papers and test on the following two dataset: Yahoo answers dataset and yelp review polarity dataset (links are below).
- You can use a smaller number of layers for the densely connected CNN.
- Feel free to train and test only on some part of the dataset if the whole dataset is too large. But please clearly mention which part of data you use in each of the dataset.

Data:
https://www.kaggle.com/soumikrakshit/yahoo-answers-dataset
https://www.kaggle.com/irustandi/yelp-review-polarity/version/1

Project 6: Forecasting Item Sales in Supermarket

How many packets of Milo will one store sell each month in a year? Forecasting sales at this level may seem as difficult as predicting the weather. Both types of forecasting rely on science and historical data. Inaccurate business forecasts could result in actual or opportunity losses. In this project, you are challenged to use appropriate methods to improve forecast accuracy for item sales. In this project, you are provided with sales data of three food items from three supermarket stores. If there are special events or holidays in a particular day, the information are also recorded. The methods used can be applied in various business areas, such as setting up appropriate inventory or service levels. It can also help reduce waste and be able to appreciate uncertainty and its risk implications. The detailed description of the dataset will be released in person.

Project 7: Dealing with severely imbalanced target variables

In many real-world applications such as fraud detection, medical diagnosis, and predictive maintenance, the datasets involved are highly imbalanced, where one class (often the event of interest) occurs far less frequently than the other. Traditional machine learning algorithms tend to be biased toward the majority class, resulting in misleading performance metrics such as high accuracy but poor detection of rare but critical events. This assignment challenges you to work

with a given dataset containing a significantly imbalanced target variable. Your task is to explore, model, and critically evaluate various strategies for handling class imbalance in a predictive modeling context. The data will be released in person.

Project 8: Neural Hawkes

Implement the Neural Hawkes model for sequential events prediction. Reproduce the results on stack overflow and the retweet dataset.

Description:

- We focus on the problem that: given a series of arrival time and types of action, we predict the next event time and type. Arrival time are generally modeled using point process models. So this project is mathematical.
- Neural Hawkes replace the intensity function of the Hawkes process with a continuous time LSTM.
- Since we have relatively small public datasets, training doesn't take too much time. CPU is enough.
- In this project, you need to use mask to deal with sequences with various length.
- Grading is based on: your accuracy on action types prediction and rmse for arrival time prediction.

Code: https://github.com/HMEIatJHU/neurawkes
Paper: https://arxiv.org/pdf/1612.09328.pdf
Data: https://drive.google.com/drive/u/0/folders/12RLDZTfsR6OF2pFJA10d6YSPFDu5YDRL

Project 9: Image to image translation using cycle GANs

Implement a Cycle-consistent Generative Adversarial Network (CycleGAN) model from scratch for image-to-image translation problems based on the paper [1].

This model can translate an input image to another image based on the model parameters. If the model parameters are trained on image collections of horses and zebras, then the model can translate an input horse image into an output zebra image in terms of color. If the model is trained on image collections of photos and art paintings, then the model can output an image in the art style of the input image. This technique is useful for paired images generation including collection style transfer, object transfiguration, season transfer, photo enhancement.

The datasets for the implementation are available at [2].

Paper: https://openaccess.thecvf.com/content_ICCV_2017/papers/Zhu_Unpaired_Image-To-Image_Translation_ICCV_2017_paper.pdf
Data: https://people.eecs.berkeley.edu/~taesung_park/CycleGAN/datasets/
Code: https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix

Another Helpful Link: https://cyclegans.github.io/

Expectations:

Successful implementation on at least one of the datasets.

- Visual results similar as in https://cyclegans.github.io/project1/2018/04/29/ Results-and-Analysis-on-Cycle-GAN-Implementation/
- Loss function values similar to Tables 1-5 in paper [1].

References:

1. Zhu, Jun-Yan, Taesung Park, Phillip Isola, and Alexei A. Efros. "Unpaired image-to-image translation using cycle-consistent adversarial networks." In Proceedings of the IEEE international conference on computer vision, pp. 2223-2232, 2017.
2. https://people.eecs.berkeley.edu/~taesung_park/CycleGAN/datasets/

Project 10: Show and Tell - A Neural Image Caption Generator

Build a generative model using latest techniques in Computer Vision and Machine Translation to describe an image. This project involves implementation of the paper [3]. In this project, a single joint model is constructed to generate a target sequence of words that describes the input image. The idea is similar to encoder-decoder RNN used in translating sentences except that a Convolutional Neural Network is used in place of the encoder RNN. CNN, which takes image as an input, is trained for an image classification task to generate a compact representation of the original image. This representation is then passed as an input to a decoder RNN that generates the sentences.

Paper: https://arxiv.org/abs/1411.4555
Data: MSCOCO 2014 [4] and Flickr30k [5]
Code: https://github.com/kelvinxu/arctic-captions

Expectations:

Implementation on at least one of the datasets and performance evaluation using BLEU, CIDEr and ROUGE L metrics similar to as follows.

Days	BLEU1	BLEU2	BLEU3	BLEU4	CIDEr	ROUGE L
COCO	64.6 | 71.3	45.9 | 54.2	31.7 | 40.7	22.0 | 27.7	69.4 | 85.5	47.6 | 53.0
Flickr30k	53.9 | 63.0	34.8 | *NA*	22.0 | *NA*	14.3 | *NA*	26.5 | *NA*	39.5 | *NA*

Table 1: There are two results in each cell. The left is for our code. The right is for the results in the paper.

References:

3. Vinyals, Oriol, Alexander Toshev, Samy Bengio, and Dumitru Erhan. ”Show and tell: A neural image caption generator.” In Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 3156-3164. 2015.

4. Chen, Xinlei, Hao Fang, Tsung-Yi Lin, Ramakrishna Vedantam, Saurabh Gupta, Piotr Doll´ar, and C. Lawrence Zitnick. ”Microsoft COCO captions: Data collection and evaluation server.” arXiv preprint arXiv:1504.00325 (2015).

5. Plummer, Bryan A., Liwei Wang, Chris M. Cervantes, Juan C. Caicedo, Julia Hockenmaier, and Svetlana Lazebnik. ”Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to sentence models.” In Proceedings of the IEEE international conference on computer vision, pp. 2641-2649. 2015.

Project 11: Physics-informed machine learning in manufacturing, and management.

[Explore]

This is an explorative project where you will need to review about the state-of-the-art of the physics-informed machine learning and find its application in your field of study.

Project 12: Explainable AI for trustworthy decision-making in manufacturing, finance and management. [Explore]

This is an explorative project where you will need to review about the state-of-the-art of the explainable AI and find its application in your field of study.